Civics Group	Index Number	Name (use BLOCK LETTERS)

H1

ST ANDREW'S JUNIOR COLLEGE
2018 JC2 Prelims

H1 BIOLOGY **8876/01**

Paper 1: Multiple Choice (Mark Scheme)

Tuesday 18th Sept 2018 1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended, not supplied)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the multiple choice answer sheet in the spaces provided.

There are **30** questions in this paper. Answer all questions. For each question, there are four possible answers, A, B, C and D.

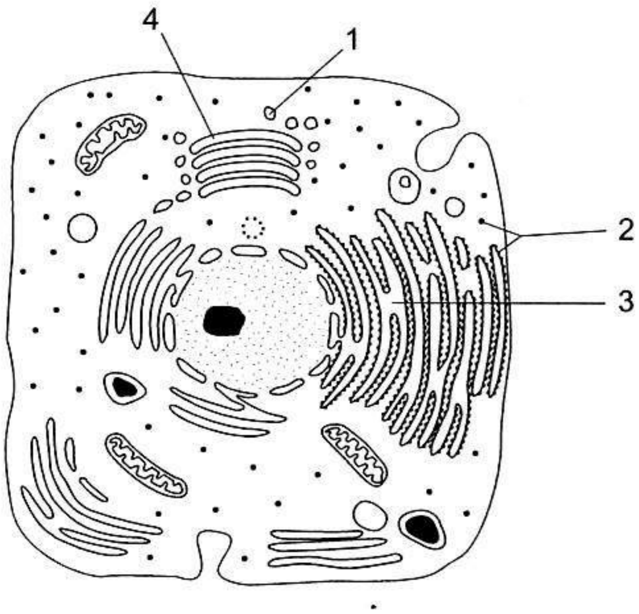
Choose the one you consider correct and record your choice in soft pencil on the separate multiple choice answer sheet.

INFORMATION TO CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for wrong answer.
Any rough working should be done in this booklet.

At the end of the examination, submit the multiple choice answer sheet and question paper separately.

1 The figure below shows the structure of an animal cell.



Which of the following correctly identifies the functions of the labelled structures?

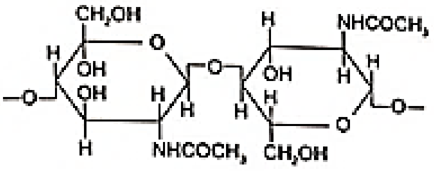
	Synthesising polypeptides from amino acids	**Transporting proteins**	**Carrying out glycosylation**	**Secreting digestive enzymes**
A	2	1	4	3
B	1	4	2	3
C	2	3	4	1
D	1	3	2	4

2 Nucleases are enzymes that hydrolyze phosphodiester bonds. In which of the following cell organelle(s) would one expect to see activity of this enzyme?

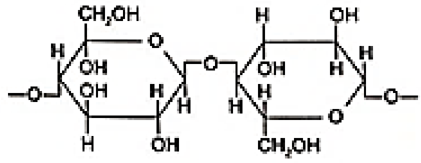
1 rough endoplasmic reticulum
2 mitochondrion
3 chloroplast
4 Golgi apparatus

A 1 only
B 1 and 4 only
C 2 and 3 only
D 1, 2 and 3 only

3 The diagrams show short sections of some common polysaccharides and modified polysaccharides.



1



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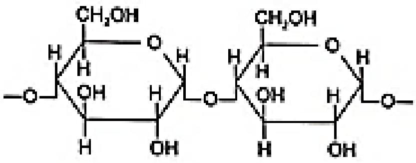

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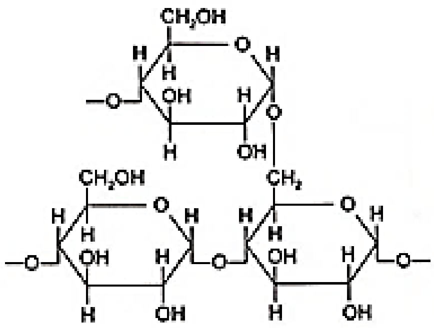

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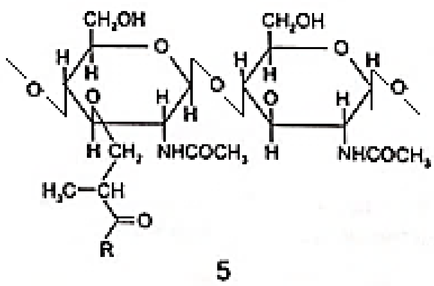


5

The polysaccharides can be described as below.

- Polysaccharide **F** is composed of β-glucose monomers with 1,4 glycosidic bonds.
- Polysaccharide **G** is composed of α-glucose monomers with 1,4 and 1,6 glycosidic bonds.
- Polysaccharide **H** is composed of N-acetylglucosamine and N-acetylmuramic acid monomers with β-1,4 glycosidic bonds.
- Polysaccharide **J** is composed of α-glucose monomers with 1,4 glycosidic bonds.
- Polysaccharide **K** is composed of N-acetylglucosamine monomers with β-1,4 glycosidic bonds.

Which shows the correct pairings of polysaccharide descriptions and diagrams?

	Polysaccharide				
	F	**G**	**H**	**J**	**K**
A	2	4	5	3	1
B	2	5	4	1	3
C	3	4	1	2	5
D	3	5	4	1	2

4 Which statement is true for phospholipid, but **not** for protein?

A Its molecules have hydrophilic and hydrophobic components.
B Its molecules are synthesized from non-identical sub-units.
C It is a barrier to polar molecules.
D It is found in cell membranes.

5 With reference to carrier proteins, which of the following statements is/are true for all carrier proteins?

1 They contain binding sites for specific molecules or ions.
2 They directly require ATP to transport substances across the membrane.
3 They are soluble globular proteins.
4 They are embedded in membranes.

A 1 only
B 1 and 4
C 3 and 4
D 1, 2 and 4

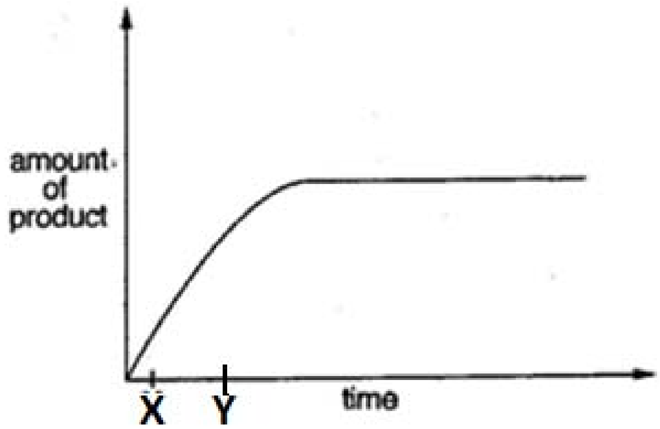
6 The list shows three characteristics of enzyme activity.

1 Reaction rate decreases if the concentration of non-competitive inhibitor increases.
2 Reaction rate is reduced at extremes of pH.
3 Reaction rate is reduced at low temperature.

What explains each of these characteristics?

	Availability of active sites is reduced	**Reduced kinetic energy reduces the rate of molecular collisions**	**Hydrogen bonding is disrupted**
A	1	2	3
B	2	1	3
C	1	3	2
D	2	3	1

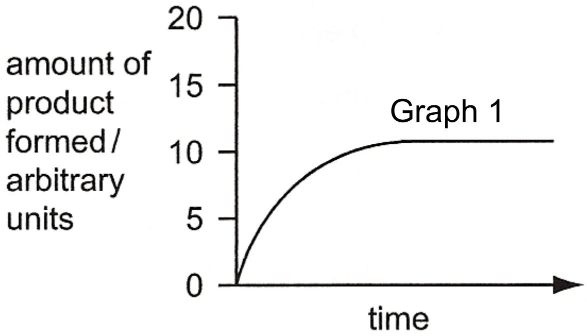
7 The graph below shows the amount of product formed in an enzyme-catalysed reaction over a certain period of time at 37°C.



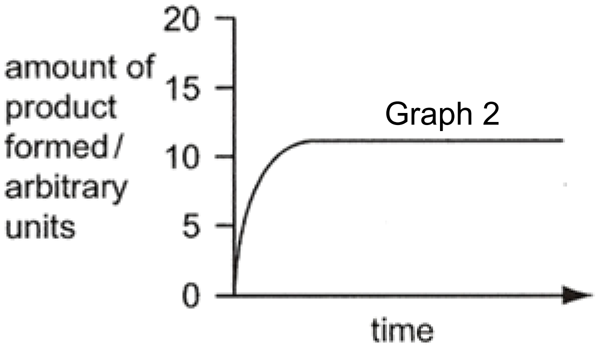
What is true at time **X**?

A Most enzyme molecules will have free active sites.
B The number of unreacted substrate molecules is high.
C The number of enzyme-substrate complexes is low.
D The rate of enzymatic reaction is lower than at time **Y**.

8 Graph 1 below shows the amount of product formed by a standard concentration of salivary enzymes and a standard concentration of substrate at a temperature of 15°C and pH of 7.



A change in condition(s) results in Graph 2.

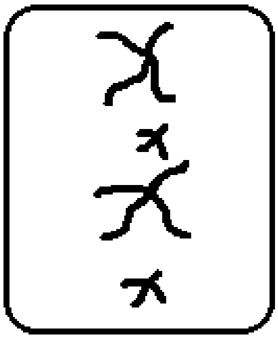


Which of the following conditions can best explain Graph 2?

1 Doubling of substrate concentration
2 Doubling of enzyme concentration
3 Increasing temperature to 20°C
4 Decreasing pH to 2

A 1 and 2
B 1 and 3
C 2 and 3
D 3 and 4

9 The diagram below shows metaphase of mitosis in a cell of an organism.



Each homologous pair of chromosomes in this organism contains 4 gene loci. This organism was genotyped and found to be heterozygous at all gene loci. The organism reproduces sexually via the production of millions of gametes by meiosis.

What is the maximum possible number of genetically different gametes that can be produced by this organism, assuming crossing over does not occur during meiosis in all cells?

A 2
B 4
C 16
D 256

10 Hybrid species can be produced from cabbage and radish.
The table below shows the chromosome numbers in the parental species and the hybrids.

type of cell	number of chromosomes per cell
parental cabbage	18
parental radish	18
parental gametes	9
F1 hybrids	18
F1 gametes	9
F2 hybrids	18
F2 gametes	18
F3 hybrids	36

Chromosomal mutation occurred at one stage. At which stage did it occur?

A during the formation of the F1 gametes.
B during the formation of the F2 gametes.
C during the fusion of the parental gametes.
D during the fusion of the F1 gametes.

11 3 different polynucleotide molecules (X, Y and Z) were isolated from a eukaryotic cell. One of them is a double-stranded DNA gene, while the other two are the pre-mRNA and mature mRNA that the DNA gene codes for.

The adenine nucleotide content of all 3 molecules was examined and shown in the table below:

Molecule	Percentage of adenine nucleotides in the molecule / %
X	49
Y	52
Z	53

Based on the information given, which of the following conclusions is/are valid and true?

1 X is definitely the DNA gene.
2 Z is definitely the mature mRNA.
3 The pre-mRNA molecule has more uracil than guanine in it.
4 Y has more purine nucleotides than pyrimidine nucleotides in it.

A 1 and 2
B 1, 2 and 4
C 2, 3 and 4
D 1, 3 and 4

12 Which of the following order of steps is true for transcription?

1 RNA polymerase II binds to promoter
2 Primase adds a RNA primer to the 3' end of template strand
3 General transcription factors recognize and bind to TATA box of promoter
4 DNA polymerase III adds complementary deoxyribonucleotides to the 3' end of the growing DNA chain
5 RNA polymerase II transcribes a DNA sequence which codes for a polyadenylation signal (AAUAAA) in the RNA transcript
6 RNA polymerase II transcribes a stop codon
7 RNA polymerase II adds complementary ribonucleotides to the 3' end of the growing RNA chain
8 DNA polymerase III binds to promoter
9 Primer is hydrolyzed and replaced by deoxyribonucleotides

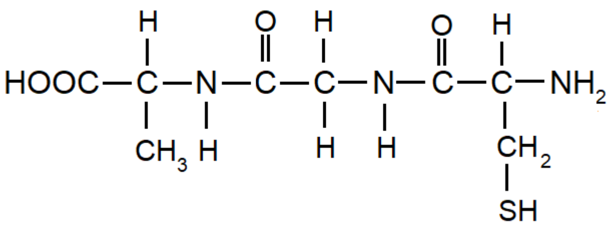
A 2, 1, 7, 5
B 2, 8, 4, 9
C 3, 1, 7, 5
D 3, 1, 7, 6

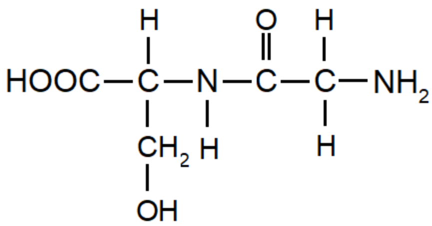
13 In an experiment, polypeptide A, which is coded for by a non-mutated version of a prokaryotic gene, was cleaved by a particular protease.

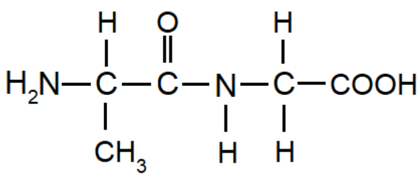
The cleavage produced 2 fragments, one of which contains the C-terminus of polypeptide A and is 5 amino acids long. This 5-amino-acid-long fragment, now called Peptide B, was then isolated for further investigation.

A solution containing many molecules of Peptide B was treated with another protease, called Protease X. The solution was analysed after the treatment and was found to contain various different fragments of different lengths and sequences.

The structures of 3 of these fragments are shown below:







It is known that Protease X is able to cleave any peptide bonds within the molecule of Peptide B. However, the cleavage of all peptide bonds within a single molecule is rare.

The mRNA codons involved in the synthesis of the Peptide B portion of Polypeptide A are shown below:

Amino acid	**R group**	**mRNA codon**
Glycine	H	5' – GGC – 3'
Alanine	CH_3	5' – GCC – 3'
Serine	CH_2OH	5' – UCC – 3'
Cysteine	CH_2SH	5' – UGU – 3'

Which of the following correctly shows a single point mutation in the portion of the template DNA sequence that codes for Peptide B, leading to a single amino acid substitution?

A 5' – GGA GCC GCC GCC ACA – 3'
B 5' – GCC GGA ACA GCC GCC – 3'
C 5' – GCC GCC ACA GGA GCC – 3'
D 5' – ACA GCC GCC GCC GGA – 3'

14 Translation did not occur successfully in a eukaryotic organism. Analysis of the structures in the cell revealed the following:

Structure	**Presence (√) / Absence (X)**
Free floating aminoacyl-tRNAs	√
Initiator tRNA- small ribosomal subunit complex	√
Initiator tRNA (base-pairing with AUG codon) in the P-site of large ribosomal subunit	√
Second aminoacyl-tRNA in the A-site of large ribosomal subunit	√
Polypeptide chain	X

Which is the most probable explanation for these observations?

A peptidyl-transferase is not functioning properly
B aminoacyl tRNA synthetases fail to attach amino acids to tRNA
C there is an error in the scanning for the start codon by tRNA-small ribosomal subunit complex
D release factor fails to hydrolyze polypeptide chain from the tRNA

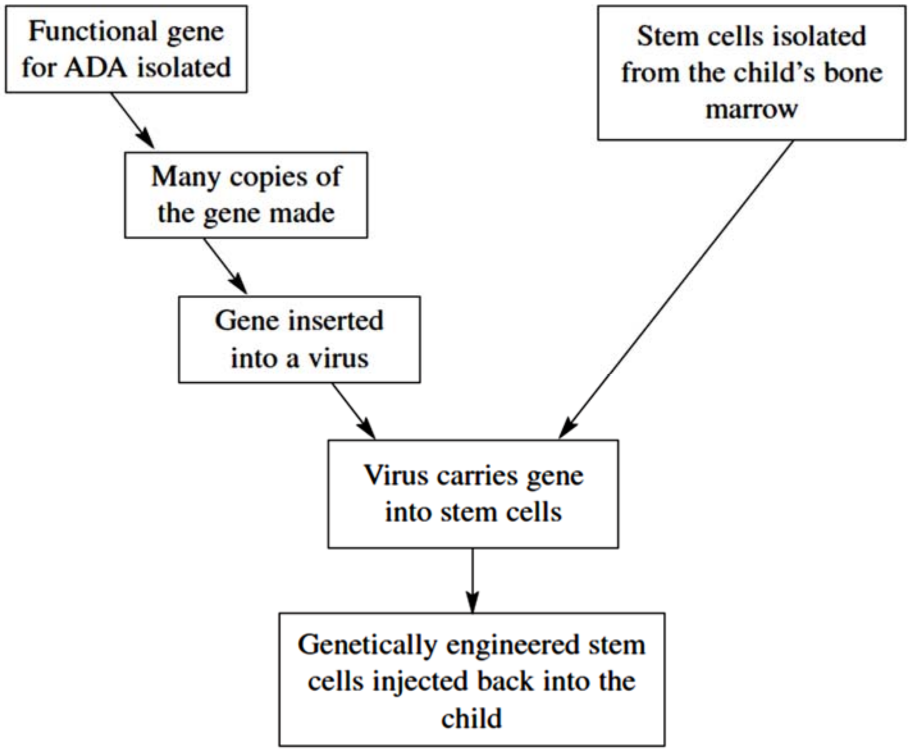
15 Which is a correct statement about obtaining human embryonic stem cells for research?

1 Removal of these cells is considered to be ethically acceptable as normal development of the embryo is not inhibited.
2 The cells must be removed at an early stage of development from a region of the blastocyst known as the inner cell mass.
3 The cells must be removed immediately following the successful fertilisation of the ovum by the sperm, and after checking for normal mitotic division.
4 The region of the blastocyst from where the cells are removed is an area that develops at a later stage into the placenta.

A 2 only
B 1 and 2
C 2 and 3
D 3 and 4

16 Children with severe combined immunodeficiency disorder (SCID) cannot produce the many types of white blood cells that fight infections. This is because they do not have the functional gene to make the enzyme ADA. Some children with SCID have been treated with stem cells.

The treatment used with the children is described in the flowchart.



Which of the following explains why stem cells can be used in the treatment of SCID?

1 They can divide mitotically to replace existing cells.
2 Due to their pluripotent nature, they have the ability to form only certain types of white blood cells that restores the ability to fight infection.
3 As the stem cells are from the child's own cells, there is no/little risk of rejection.
4 They possess a unique set of genome to allow for multipotency.

A 1 and 2
B 1 and 3
C 2 and 4
D 3 and 4

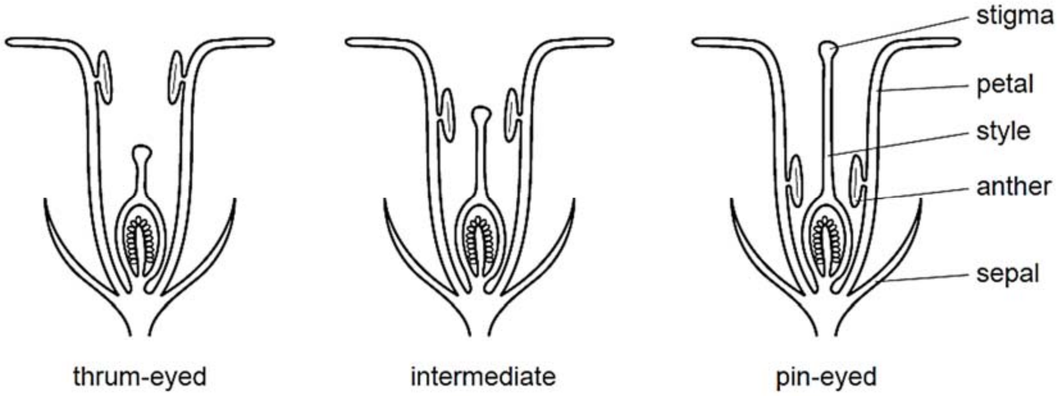
17 The primrose, *Primula vulgaris*, is a small herbaceous, yellow-flowered plant which is common in cooler areas of the Northern hemisphere including alpine and Arctic areas.

The flowers of the primrose have different flower shapes (polymorphic), which are adaptations for pollination. 'Thrum-eyed' primroses have a short style. 'Pin-eyed' primroses have much longer styles. The anther position also varies among the primrose.

Some populations of primrose consist almost entirely of plants with intermediate flowers. These populations are common where there are fewer winged insects.

Anthers produce pollen (male gametes) which land on the stigma, leading to fertilization.

The diagrams show polymorphic flowers of primroses.



Which statements are correct?

1 Cross-pollination will be favoured between pin-eyed and thrum-eyed primroses.
2 Primroses with pin-eyed flowers are likely to show more genetic diversity than primroses with intermediate flowers.
3 Primroses with thrum-eyed flowers are likely to be more able to adapt to changing environmental conditions than pin-eyed primroses.
4 Self-pollination is more likely to occur in primroses with intermediate flowers.

A 1 and 2
B 3 and 4
C 1, 2 and 4
D All of the above

18 On the tiny Lord Howe Island, 600 miles east of Australia, there are two species of palm which seem, from DNA analysis, to be descended from one original species. Factors involved in this speciation on this tiny island include:

1 linkage of genes for soil tolerance and flowering time
2 variation in flowering time
3 variation of soil tolerance
4 variation of soil types on the island

What is the correct sequence to explain this speciation?

A $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$
B $2 \rightarrow 1 \rightarrow 4 \rightarrow 3$
C $3 \rightarrow 4 \rightarrow 1 \rightarrow 2$
D $4 \rightarrow 3 \rightarrow 2 \rightarrow 1$

19 A large population of equal numbers of dark and light mice was released into an area where owls are predators of mice. Because of predation, only 25 % of these mice survived and selective killing by owls changed the proportions of dark to light mice from 1:1 to 4:1.

If mice produce an average of eight offspring per litter and the pattern of predation remains the same, what would happen to the population of light mice?

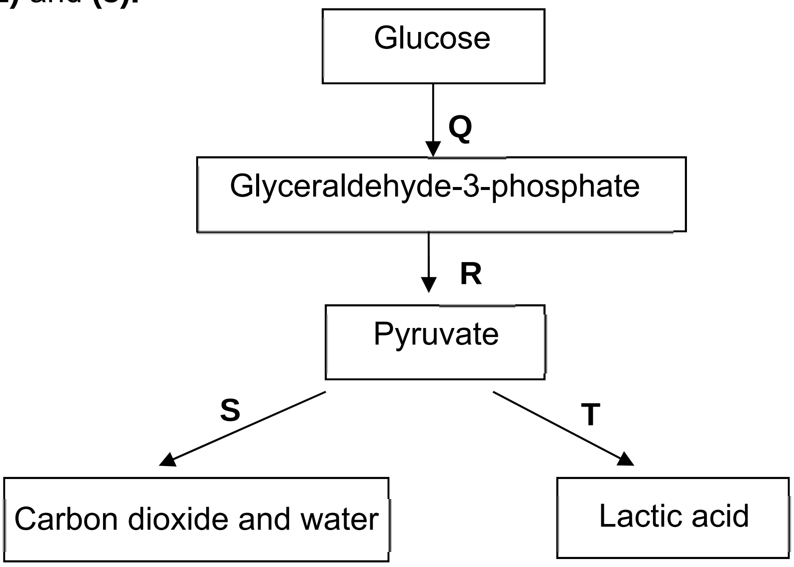
A They disappear after one further generation.
B They disappear after two further generations.
C They remain at the level of one fifth of the population.
D They will be reduced to a constant but very low frequency.

20 The frequency of recessive alleles in population is influenced by selection pressure.

Which row shows the conditions in which recessive alleles are retained in a population?

	Environmental variation of habitats	Heterozygote advantage	No selective advantage	Polymorphism (many phenotypes)
A	✓	✓	✓	✓
B	✓	×	×	✓
C	×	✓	×	×
D	×	×	✓	×

21 With reference to the diagram below, relate processes **Q, R, S, T** to statements **(1), (2)** and **(3).**



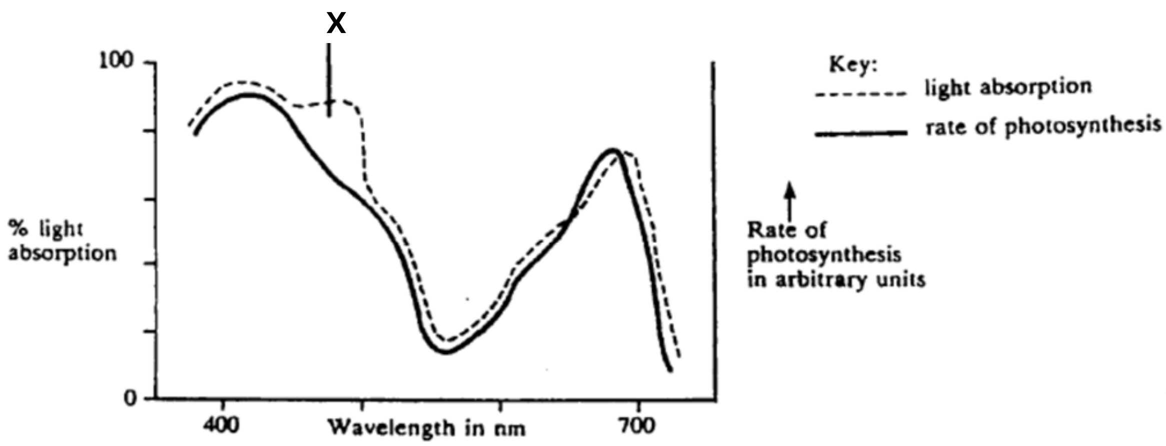
(1) NAD is regenerated without the use of the electron transport system
(2) ATP is synthesised via substrate level phosphorylation
(3) It can take place under anaerobic conditions.

	(1)	**(2)**	**(3)**
A	T only	R only	Q,R,T only
B	T only	R,S only	Q,R,T only
C	S,T only	R only	Q,R,S,T
D	S,T only	R,S only	Q,R,S,T

22 A major function of the mitochondrial inner membrane is the conversion of energy from electrons to the stored energy of the phosphate bond in ATP. To accomplish this function, this membrane must have all of the following features except

A high permeability to protons.
B integral, transverse ATP synthase.
C proteins to accept electrons from NADH.
D proton pumps embedded in the membrane.

23 The graph below shows the effect of different wavelengths of light on the rate of photosynthesis and on the amount of light absorbed by the pigments in a green seaweed.



The difference between the two curves at X is due to

A inefficient trapping of light energy by the chlorophyll
B no ATP production at that wavelength
C oxygen given off during photosynthesis interferes with the absorption of light.
D carotenes absorbing light that is not used in photosynthesis.

24 The incidence of primaquine sensitivity in a population is 14% in men and 2% in women. A proportion of the remaining women, but not the men, exhibit mild sensitivity.

What does this suggest about the inheritance of this defect?

A autosomal, codominant
B autosomal, recessive
C sex-linked, codominant
D sex-linked, recessive

25 In a series of plant breeding experiments, a pure-breeding plant with big and hairy leaves was crossed with a pure-breeding plant with small and hair-less leaves. The leaves in the F_1 generation were all big and hairy. Self-fertilisation of the F_1 generation produced the following results:

905	big and hairy leaves
301	big and hair-less leaves
305	small and hairy leaves
98	small and hair-less leaves

An F_2 plant with big and hairy leaves was crossed with an F_2 plant with small and hairy leaves. What is the maximum proportion of plants with small and hair-less leaves that could have appeared in the resulting progeny?

A 0%
B 12.5%
C 25%
D 50%

26 Fruit flies *Drosophila* homozygous for long wings, were crossed with flies homozygous for vestigial wings. The F_1 and F_2 generations were raised at three different temperatures.

At each temperature, the F_1 generation all had long wings.

The table below shows the results in the F_2 generation.

Temperature	***Result***
21°C	¾ long wings, ¼ vestigial wings
26°C	¾ long wings, ¼ intermediate wing length
31°C	all long wings

Which statement explains these results?

A Wing length is under polygenic control.
B Long wing and vestigial wing illustrate codominance at 26°C.
C Heterozygous flies have vestigial wings only at 21°C or below but have long wings at 31°C or above.
D Vestigial wing is recessive but causes a vestigial wing phenotype only at lower temperatures.

27 The table shows the results of a study made on a large number of twins.

Twin group	Mean difference in eye colour intensity/a.u.	Mean difference in weight/kg
Identical, raised together	1.7	2.0
Identical, raised apart.	1.8	4.8
Non-identical, same-sex, raised together	4.4	4.9

What do these results suggest about the influence of genes and environment on eye colour intensity and weight in humans?

A Genes have a greater influence than the environment on the eye colour intensity and the weight of identical twins.
B Eye colour intensity and weight are influenced by the environment.
C Weight is influenced by environment and genes; eye colour intensity is mainly influenced by genes.
D The environment has more effect than genes on the eye colour intensity and weight of non-identical twins.

28 The length of the petiole (leaf stalk) in a type of flowering plant is controlled by two genes, A and B. These genes are found on different loci on non-homologous chromosomes.

Homozygous dominant plants have long petioles (30 cm), homozygous recessive plants have short petioles (10 cm). Each dominant allele contributes 5cm to the petiole length.

F_1 plants with medium length petioles (20 cm) were obtained when a plant with short petiole is crossed with a plant with long petiole. If the F1 generation plants were allowed to cross, what proportion of their offspring would be expected to have medium length (20 cm) petioles?

A 0.0625
B 0.25
C 0.375
D 0.5

29 The Himalayan rabbits have white hair on the body and black hair on the extremities such as feet, tail, ears and face.

The allele for the Himalayan rabbit pigment pattern, $\mathbf{c^h}$, is recessive to the alleles for normal colour (all hair agouti), **C**, as well as dark chinchilla (all hair dark grey), $\mathbf{c^{chd}}$, and is dominant to the allele for albino (all hair white, no pigment production), **c**.

All of the alleles of this gene produce different versions of the same enzyme involved in pigment production.

A patch of white fur was removed from a Himalayan rabbit and an ice pack secured to the skin. The fur that grew back on the patch was black.

Which is correct?

	Genotypes of Himalayan rabbits	**Explanation for pigment pattern in Himalayan rabbits**
A	c^hc^h only	The enzyme is denatured at the high skin temperatures found on the rabbit's bodies
B	c^hc^h only	The enzyme becomes inactive at the low skin temperatures found on the rabbit's feet, tail, ears and face.
C	c^hc^h and c^hc only	The enzyme is denatured at the high skin temperatures found on the rabbit's bodies
D	c^hc^h and c^hc only	The enzyme becomes inactive at the low skin temperatures found on the rabbit's feet, tail, ears and face.

30 The Southern pine beetle is a pest native to pine forests in Central America and the southeastern U.S..

However recent observations show the latitude of this pest infestation creeping northward by about 40 miles a decade since 1980, and could damage 273,000 square miles of pine forests by 2080.

Which of the following explanations for the above observation are attributed to climate change?

1 Longer and more intense droughts weakening the defenses of trees, making them vulnerable to attack by the beetles.
2 Long-term suppression of forest fires leaving pine forests unnaturally dense and uniform, facilitating the beetles' spread from tree to tree.
3 Pines trees colonising new territories with cooler climates.
4 Increased temperatures in the winter allowing the beetles' lavae to survive.

A 1 and 4 only
B 2 and 3 only
C 1, 3 and 4 only
D All of the above

\- **END OF PAPER** -